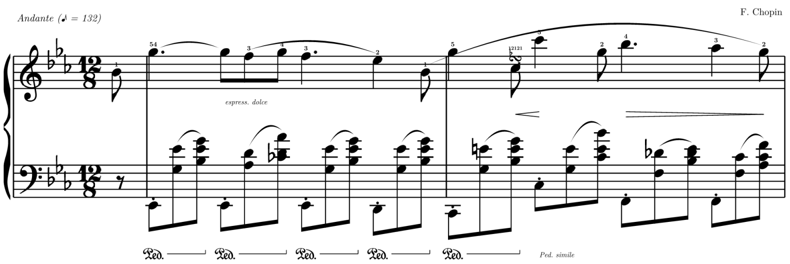
typed-scores

Western music notation, natively in Typst

User Guide



Version 0.1.0

Contents

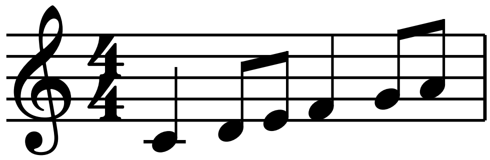
1 Introduction

`typed-scores` turns compact musical text into engraved notation inside Typst. The Rust/WASM layer parses pitches and exact durations; the Typst layer aligns voices, packs systems, and draws bundled Bravura glyphs through CeTZ. No music font installation or external engraving program is required.

Use the package for theory worksheets, analytical examples, lecture notes, short compositions, and piano or ensemble excerpts where musical content should remain readable and versionable as text.

Typst

```
#bar(
  "C4:q D4:e E4:e F4:q G4:e A4:e",
  clef: "treble",
  time: "4/4",
  beams: true,
)
```



The current release supports notes through thirty-seconds, two dots, chords, rests, ties, automatic rests, beat-aware beams, accidentals, four clefs, key and time signatures, named slurs, fingering, combined articulations, text directions, turns, hairpins, pedal spans, pickup measures, grand staves, ensemble staves, and automatic system wrapping.

2 Getting started

2.1 Installation and import

Import every public symbol from the Typst preview namespace:

```typst
#import "@preview/typed-scores:0.1.0": *
```

Or import only the builders you use:

```typst
#import "@preview/typed-scores:0.1.0": note, bar, score
```

The public rendering API is:

Symbol	Purpose
`note`	Render one note, chord, or rest on a staff.
`bar`	Render and optionally validate one measure.
`score`	Render one or more aligned staves across measures and systems.
`grand-staff`	Piano-oriented wrapper around `score` .
`staff`	Compatibility wrapper for a single rendered event.

> **Note.** The low-level drawing functions from `render.typ` are also exported. They are useful for custom extensions, but `note` , `bar` , and `score` are the stable user-facing entry points.

2.2 Your first score

```typst
#score(
  treble: "C5:q D5:q E5:q F5:q",
  bass: "C3:h G2:h",
  key: "C",
  time: "4/4",
)
```

The strings contain musical events. The score shares every onset across its staves, so notes beginning together receive the same horizontal coordinate.

3 Note language

3.1 Pitches

Pitches use an uppercase letter, optional accidental, and scientific octave:

Text	Pitch
`C4`	Middle C.
`F#5`	F-sharp in octave 5.
`Bb2`	B-flat in octave 2.
`E4`	E-natural. In a flat key this prints a natural sign when needed.

The supported clefs are `treble` , `bass` , `alto` , and `tenor` . Ledger lines are generated from the diatonic staff position.

3.2 Durations and dots

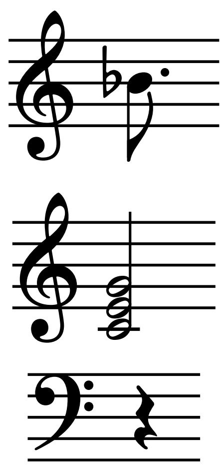
Code	Value	Code	Value
`w`	Whole	`w.` / `w..`	Dotted / double-dotted whole
`h`	Half	`h.` / `h..`	Dotted / double-dotted half
`q`	Quarter	`q.` / `q..`	Dotted / double-dotted quarter
`e`	Eighth	`e.` / `e..`	Dotted / double-dotted eighth
`s`	Sixteenth	`s.` / `s..`	Dotted / double-dotted sixteenth
`t`	Thirty-second	`t.` / `t..`	Dotted / double-dotted thirty-second

Join pitch and duration with a colon (`Eb4:q`) or use the compact form (`Eb4q`). The two spellings are identical.

3.3 Notes, chords, and rests

Typst

```
#note("Bb4:e.", clef: "treble")
#note("(C4 E4 G4):h", clef: "treble")
#note("r:q", clef: "bass")
```

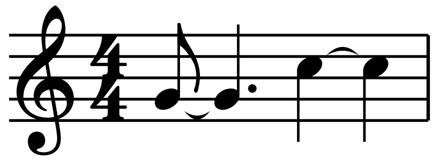


Chord pitches are space-separated inside parentheses and share one duration. Rests begin with `r:` and use the same duration codes.

3.4 Ties

Put `~` after the source event. A chord tie produces one curve per pitch.

```typst
#bar(
  "G4:e ~ G4:q. C5:q ~ C5:q",
  time: "4/4",
)
```

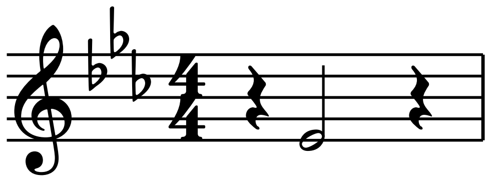


Within a system, the tie goes to the next event. At a system edge it continues to the margin and resumes on the next system.

3.5 Automatic rests

An underscore is a rest placeholder. When a time signature is known, the remaining duration is divided across all placeholders using supported note values.

```typst
#bar(
  "_ Eb4:h _",
  key: "Eb",
  time: "4/4",
)
```

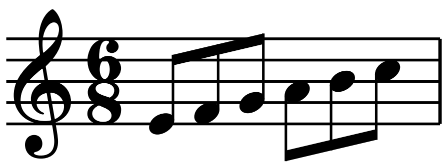


If the remaining value cannot be represented across the requested number of placeholders, compilation fails with the remaining rational duration.

3.6 Beams and manual breaks

Set `beams: true` to connect flagged notes. Groups break at rests, manual `/` markers, and beat boundaries. Compound meters such as 6/8, 9/8, and 12/8 use dotted beats.

```typst
#bar(
  "E4:e F4:e G4:e / A4:e B4:e C5:e",
  time: "6/8",
  beams: true,
)
```



4 Expressive notation

Annotations live in square brackets after an event. They do not change the event duration. Put spaces between multiple annotations.

```
G5:q.[f=4 s1()
```

Typst

4.1 Annotation reference

Annotation	Meaning
`f=4`	Fingering outside the articulation stack on its placement side.
`stacc`	Staccato dot.
`staccatissimo`	Wedge staccatissimo.
`tenuto` / `legato`	Tenuto line.
`accent`	Accent.
`marcato` / `strong`	Marcato.
`text=espress._dolce`	Italic direction near the staff; underscores become spaces.
`text-below=Ped._simile`	Italic direction below low beams and pedal marks.
`turn`	Turn ornament.
`chromatic-turn`	Turn with flat-above and natural-below auxiliaries.
`turn-f=12121`	Fingering shown above a turn.
`s1(` / `s1)`	Open / close named slur `s1` .
`p1(` / `p1)`	Open / close pedal span `p1` .
`h1<` / `h1!`	Open crescendo / stop hairpin `h1` .
`h1>` / `h1!`	Open diminuendo / stop hairpin `h1` .

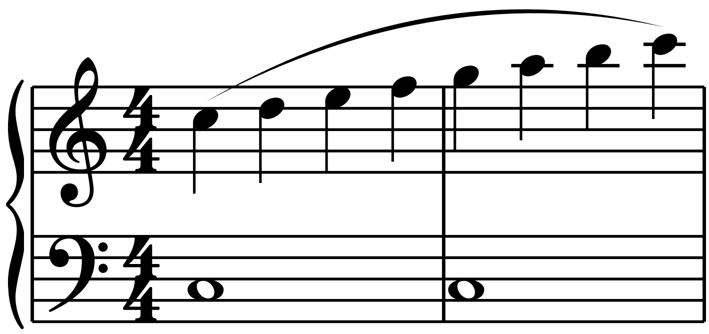
IDs pair the ends of a spanner. They may contain letters and digits. A slur ID must begin with `s` , a pedal ID with `p` , and a hairpin ID with `h` . Use distinct IDs for overlapping spans.

Articulations combine by listing multiple names, such as `[marcato stacc]` , `[tenuto staccatissimo]` , or `[accent tenuto]` . Duration marks sit nearest the notehead and force accents outside them. The entire stack follows the notehead side of the effective stem: above stem-down notes and below stem-up notes, including beamed groups.

4.2 Slurs

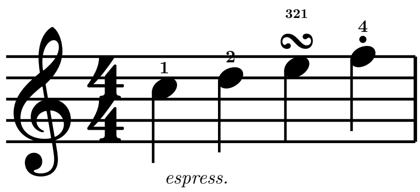
Named slurs may stay within a measure, cross barlines, nest, or overlap. Slurs are placed above their anchor notes and raised to clear interior note anchors.

Typst
```
#score(
  treble: (
    "C5:q[s1(] D5:q E5:q F5:q",
    "G5:q A5:q B5:q C6:q[s1)]",
  ),
  bass: ("C3:w", "C3:w"),
  time: "4/4",
)
```



4.3 Fingering, text, and ornaments

Typst
```
#bar(
  "C5:q[f=1 text=espress.] D5:q[f=2] E5:q[turn turn-f=321] F5:q[stacc f=4]",
  time: "4/4",
)
```



4.4 Hairpins and pedal spans

Hairpins attach to event onsets. Pedal spans print a Bravura Ped. mark and a release bracket.

Typst

```
#score(
  treble: "C5:q[h1<] D5:q E5:q F5:q[h1!]",
  bass: "C3:e[p1(] G3:e E4:e[p1)] C3:e[p2(] G3:e E4:e[p2)] r:q",
  time: "4/4",
  beams: true,
  staff-gap: 11,
)
```

5 Single-staff builders

5.1 `note()`

`note` draws one event with a clef and enough staff width for its ink.

Argument	Type	Default	Description
`note-str`	`str`	required	One note, chord, or rest event.
`clef`	`str`	`"treble"`	`"treble"` , `"bass"` , `"alto"` , or `"tenor"` .
`width`	`number | none`	`none`	Staff width in staff spaces; automatic when omitted.
`scale`	`number`	`1.0`	Uniform score scale. One staff space is `8pt * scale` .

5.2 `bar()`

`bar` parses a sequence, validates its duration when `time` is set, and draws a single staff measure.

Argument	Type	Default	Description
`sequence-str`	`str`	required	Space-separated event sequence.
`clef`	`str`	`"treble"`	Clef name.
`width`	`number | none`	`none`	Explicit measure width in staff spaces.
`scale`	`number`	`1.0`	Uniform score scale.
`note-spacing`	`number`	`3.1`	Base rhythmic spacing in staff spaces.
`beams`	`bool`	`false`	Draw automatic beam groups.
`key`	`str`	`"C"`	Major or minor key name.
`time`	`str | none`	`none`	Time signature and expected duration, such as `"4/4"` .

5.3 `staff()`

`staff` is a compatibility wrapper around `note` . Pass its event in `note-str:` . The `body` form is reserved for a future structured DSL.

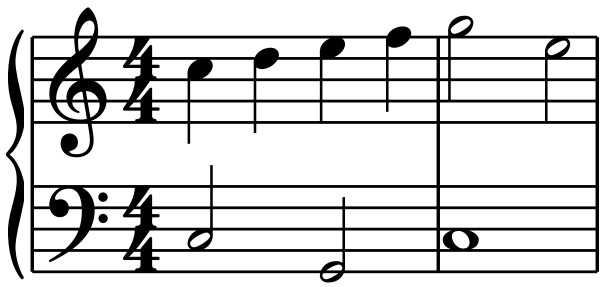
6 Scores and systems

6.1 Parallel treble and bass

For a piano score, pass a string for one measure or arrays for multiple measures. Parallel arrays must have equal lengths.

Typst

```
#score(
  treble: (
    "C5:q D5:q E5:q F5:q",
    "G5:h E5:h",
  ),
  bass: (
    "C3:h G2:h",
    "C3:w",
  ),
  time: "4/4",
)
```



An array item may be a dictionary with `notes` , `key` , `time` , or `partial` fields. Key and meter changes found on one voice apply to the whole measure; conflicting changes are rejected.

6.2 Measure dictionaries

The `bars:` form keeps all measure-wide properties together.

```typst
#score(
  bars: (
    (
      key: "Eb",
      time: "12/8",
      partial: "1/8",
      treble: "Bb4:e",
      bass: "r:e",
    ),
    (
      treble: "G5:q. F5:e G5:e Bb5:e Ab5:q. G5:q F5:e",
      bass: "Eb2:e (G3 Eb4):e (Bb3 Eb4 G4):e Eb2:e (Ab3 D4):e (Cb4 D4 Ab4):e Eb2:e (G3 Eb4):e (Bb3 Eb4 G4):e D2:e (G3 Eb4):e (Bb3 Eb4 G4):e",
    ),
  ),
  beams: true,
)
```

Measure dictionary keys

Key	Type	Default	Description
`key`	`str`	inherits	Key signature from this measure onward.
`time`	`str`	inherits	Time signature from this measure onward.
`partial`	`str | none`	`none`	Exact duration for an incomplete measure, such as `"1/8"` .
`treble`	`str`	—	Treble sequence; pair with `bass` .
`bass`	`str`	—	Bass sequence; pair with `treble` .
`voices`	`array`	—	Explicit voice dictionaries instead of treble/bass.

6.3 Pickup measures

`partial` changes validation, not the displayed meter. In the example above, both staves must equal 1/8 while the prologue still prints 12/8. The next measure returns to the inherited full 12/8 duration.

6.4 Voice arrays

Use `voices:` for ensembles or non-piano clef combinations.

```typst
#score(
  voices: (
    (name: "Violin", clef: "treble", notes: (
      "E5:q F5:q G5:q A5:q",
      "G5:h E5:h",
    )),
    (name: "Viola", clef: "alto", notes: (
      "C4:q D4:q E4:q F4:q",
      "E4:h C4:h",
    )),
    (name: "Cello", clef: "bass", notes: (
      "C3:h G2:h",
      "C3:w",
    )),
  ),
  time: "4/4",
)
```

Voice dictionary keys

Key	Type	Default	Description
`name`	`str`	generated	Stable identity used in diagnostics and slur validation.
`clef`	`str`	`"treble"`	Voice clef.
`notes`	`str | array`	required	One measure or an array of measure strings/dictionaries.

Every measure must keep the same voice count, names, and clefs. Start a new `score` call when the ensemble itself changes.

6.5 Sections

Sections carry new defaults across a run of measures. A section accepts `key` , `time` , `tempo` , and one of `voices` , `bars` , or parallel `treble` / `bass` arrays.

```typst
#score(
  sections: (
    (
      key: "C",
      time: "4/4",
      tempo: "Moderato",
      treble: ("C5:q D5:q E5:q F5:q", "G5:w"),
      bass: ("C3:h G2:h", "C3:w"),
    ),
    (
      key: "Cm",
      tempo: "Meno mosso",
      treble: ("Eb5:q D5:q C5:q Bb4:q", "C5:w"),
      bass: ("C3:h G2:h", "C3:w"),
    ),
  ),
)
```

6.6 `score()` argument reference

Argument	Type	Default	Description
`body`	`content | none`	`none`	Reserved structured-score body.
`note-str`	`str`	`"C4:q"`	Fallback event when no multi-staff input is supplied.
`clef`	`str`	`"treble"`	Clef for the fallback event.
`voices`	`array | none`	`none`	Explicit stable voice definitions.
`sections`	`array | none`	`none`	Section dictionaries with persistent defaults.
`treble`	`str | array | none`	`none`	Treble measure(s).
`bass`	`str | array | none`	`none`	Bass measure(s).
`bars`	`array | none`	`none`	Measure dictionaries.
`key`	`str`	`"C"`	Initial key. Minor names use `m` , such as `"Cm"` .
`time`	`str`	`"4/4"`	Initial meter and default measure duration.
`tempo`	`str | content | none`	`none`	Tempo text above the first system or section.
`bpm`	`number | none`	`none`	Numeric suffix combined with `tempo` when both are set.
`composer`	`str | content | none`	`none`	Composer credit above the first system, right aligned.

`width`	`number | none`	`none`	System packing width in staff spaces; available layout width when omitted.
`scale`	`number`	`1.0`	Uniform staff-space scale.
`note-spacing`	`number`	`3.1`	Base horizontal rhythmic spacing.
`beams`	`bool`	`false`	Draw automatic beam groups.
`staff-gap`	`number | none`	`none`	Fixed bottom-line distance between adjacent staves.
`wrap`	`bool`	`true`	Pack measures into multiple systems.
`system-gap`	`length`	`1.2em`	Vertical Typst space between wrapped systems.

6.7 `grand-staff()`

`grand-staff` forwards piano-oriented arguments to `score` . It accepts `treble` , `bass` , or `bars` plus the same key, meter, spacing, scale, wrapping, staff-gap, and composer options. Use `score` when you need `voices` , `sections` , `tempo` , or `bpm` .

6.8 System packing

With `wrap: true` , each measure is packed atomically into the available width. Every new system redraws clefs and the active key and time signatures. Set `width` for predictable package-level packing independent of the surrounding column width. Set `wrap: false` for an intentionally unbroken line such as the Chopin release fixture.

7 Engraving behavior

7.1 Exact durations and onset alignment

Durations are stored as reduced rational fractions of a whole note. The WASM plugin computes every event onset exactly; the Typst layout merges those onsets across voices before assigning x coordinates. A half note and two quarters therefore align without floating-point drift.

7.2 Key signatures and accidental state

Major keys from C-flat through C-sharp are supported. Minor key names map to their relative-major signatures. A matching accidental is suppressed by the key signature. A conflicting accidental prints a sharp, flat, or natural and persists for that letter and octave until the barline.

7.3 Rhythmic spacing

`note-spacing` is the base quarter-note advance. Longer values receive more room sub-linearly; shorter values are clamped so noteheads, flags, dots, and accidentals remain clear. All staves share the strictest spacing demand at each onset.

7.4 Stems and beams

Single-event stems are chosen from the average staff position. Beam groups use all pitches in the group, cap their slope, and enforce minimum stem lengths. Mixed sixteenth/eighth groups receive full beam segments and local beam stubs.

7.5 Curves

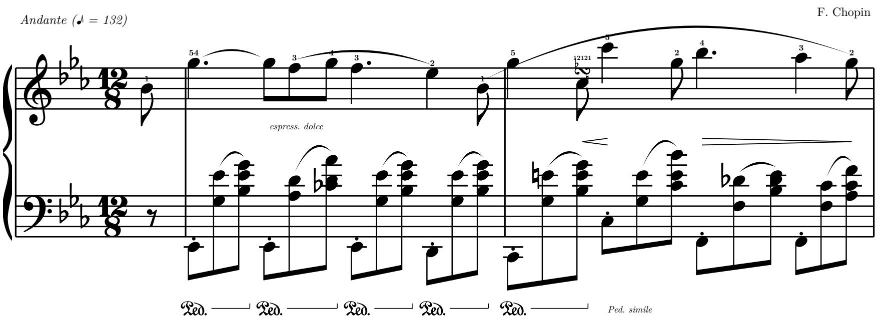
Tie curves go opposite the stem side. Slurs sit above their note anchors and grow to clear interior events. Each edge is one exact quadratic parabola, `y(t) = (1-t)sy + t*ey + 4h*t*(1-t)` , converted exactly to CeTZ's cubic path representation. There is no midpoint join or piecewise change in curvature.

7.6 Bravura assets

Clefs, noteheads, rests, flags, time-signature digits, accidentals, articulations, turns, pedal marks, braces, and brackets are SVGs extracted from Steinberg's Bravura SMuFL font. Geometry follows the bundled metadata and one canvas unit always equals one staff space.

8 Worked example: Chopin

The release-gate fixture is the opening pickup and measures 1–2 of Frédéric Chopin's Nocturne in E-flat major, Op. 9 No. 2. It exercises the complete first-release path: 12/8 compound beaming, an exact 1/8 pickup, key-signature accidentals and E-natural cancellation, tied melody, local and long slurs, fingerings, a chromatic turn, hairpins, low piano accompaniment, and pedals.



The canonical Typst source is `examples/chopin-opening.typ` . It is imported by the visual test and by the README image generator, so all three artifacts render the same musical data.

```typst
#import "../src/lib.typ": score

// Opening pickup and measures 1–2 of Chopin's Nocturne in E-flat major,
// Op. 9 No. 2, following the Mutopia transcription of the 1881 Schirmer score.
#let chopin-opening(
  scale: 0.62,
  note-spacing: 4.0,
  staff-gap: 13,
  composer: [F. Chopin],
) = score(
  bars: (
    (
      partial: "1/8",
      treble: "Bb4:e[f=1]",
      bass: "r:e",
    ),
    (
      treble: "G5:q.[f=54] ~ G5:e[text=espress._dolce] F5:e[f=3 s1() G5:e[f=4] F5:q.[f=3] Eb5:q[f=2 s1)] Bb4:e[f=1 s2()",
      bass: "Eb2:e[stacc p1() (G3 Eb4):e[sb1() (Bb3 Eb4 G4):e[sb1) p1)] Eb2:e[stacc p2() (Ab3 D4):e[sb2() (Cb4 D4 Ab4):e[sb2) p2)] Eb2:e[stacc p3() (G3 Eb4):e[sb3() (Bb3 Eb4 G4):e[sb3) p3)] D2:e[stacc p4() (G3 Eb4):e[sb4() (Bb3 Eb4 G4):e[sb4) p4)]",
    ),
    (
      treble: "G5:q[f=5] C5:e[chromatic-turn turn-f=12121 h1<] C6:q[f=5 h1!] G5:e[f=2] Bb5:q.[f=4 h2>] Ab5:q[f=3] G5:e[f=2 s2) h2!]",
      bass: "C2:e[stacc p5() (G3 E4):e[sb5() (Bb3 E4 G4):e[sb5) p5)] C3:e[stacc text-below=Ped._simile] (G3 E4):e[sb6() (C4 E4 Bb4):e[sb6)] F2:e[stacc] (F3 Db4):e[sb7() (Bb3 Db4 E4):e[sb7)] F2:e[stacc] (F3 C4):e[sb8() (Ab3 C4 F4):e[sb8)]",
    ),
  ),
  key: "Eb",
  time: "12/8",
  tempo: [Andante (♪ = 132)],
  composer: composer,
  scale: scale,
  note-spacing: note-spacing,
  staff-gap: staff-gap,
  beams: true,
  wrap: false,
)
```

The pitches, rhythms, and historical score text were checked against the public-domain 1881 G. Schirmer score from the Library of Congress and the Mutopia transcription made from that scan.

9 Diagnostics and validation

Errors begin with `typed-scores error` when they come from score-level validation. Diagnostics identify the voice and one-based bar number whenever possible.

Condition	Result
Malformed pitch or duration	Parser reports the invalid character or missing component.
Wrong full-measure duration	Actual reduced duration and expected time signature.
Wrong pickup duration	Actual reduced duration and declared `partial` value.
Unequal parallel arrays	Voice name, actual bar count, and expected bar count.
Voice structure changes	Bar where names or clefs stop matching the first measure.
Invalid slur lifecycle	Named slur and the bar where it opens or closes incorrectly.
Impossible automatic rests	Remaining rational duration and placeholder count.

Typical failures can be tested as source snippets without compiling them into the guide:

```typst
#score(treble: ("C5:w", "D5:w"), bass: ("C3:w",))
#score(treble: "C5:q", bass: "C3:w", time: "4/4")
#score(treble: "C5:q[s1(] D5:q E5:q F5:q", bass: "C3:w")
```

10 Limitations

- One rhythmic voice per staff is supported. Simultaneous independent voices on one staff are not yet modeled; use separate staves when that is musically acceptable.
- Grace notes, tuplets, tremolos, arpeggio signs, lyrics, dynamics glyphs, repeat structures, volta brackets, and specialized final barlines are not yet in the public DSL.
- Enharmonic spellings support one sharp or flat. Double accidentals are not parsed.
- Pedals and hairpins are event-anchored. They do not split automatically at a system break yet.
- Dense editorial markings may require manual `staff-gap` , `note-spacing` , or `scale` adjustments.
- The exported `voice` and `pedal` function names are reserved for a future structured body DSL. Current multi-staff voices and pedal spans use score dictionaries and annotations.

11 Quick reference

11.1 Builders

<table>
<tr><th>Call</th><th>Result</th></tr>
<tr><td><code>note(note-str, clef:, width:, scale:)</code></td><td>One event on a clefed staff.</td></tr>
<tr><td><code>bar(sequence-str, clef:, width:, scale:, note-spacing:, beams:, key:, time:)</code></td><td>One validated measure.</td></tr>
<tr><td><code>score(voices: | sections: | treble:/bass: | bars:, ...)</code></td><td>Aligned multi-measure score with wrapping.</td></tr>
<tr><td><code>grand-staff(treble:, bass:, bars:, ...)</code></td><td>Piano wrapper around <code>score</code> .</td></tr>
<tr><td><code>staff(note-str:, clef:, width:, scale:)</code></td><td>Compatibility wrapper around <code>note</code> .</td></tr>
</table>

11.2 Event grammar

Syntax	Meaning
`C4:q` / `C4q`	Single note.
`(C4 E4 G4):h`	Chord.
`r:q`	Rest.
`_`	Automatic rest placeholder.
`~`	Tie the preceding event.
`/`	Break beam before the next event.
`[annotation ...]`	One or more expressive annotations.

11.3 Score dictionary keys

Context	Keys
Measure	`key, time, partial, treble, bass, voices`
Voice	`name, clef, notes`
Section	`key, time, tempo, voices, bars, treble, bass`

11.4 Annotation summary

Syntax	Mark
`f=text`	Fingering
`stacc` / `staccatissimo`	Staccato / wedge staccatissimo
`tenuto` / `legato`	Tenuto
`accent`	Accent
`marcato` / `strong`	Marcato
`text=...` / `text-below=...`	Italic directions

`turn` / `chromatic-turn`	Turn ornament
`turn-f=text`	Turn fingering
`sID(` … `sID)`	Slur
`pID(` … `pID)`	Pedal span
`hID<` or `hID>` … `hID!`	Hairpin

12 License and credits

Package code is released under the MIT License. Bundled Bravura glyphs are released under the SIL Open Font License 1.1. The Chopin composition and the 1881 Schirmer score used for the worked example are public domain.

`typed-scores` uses CeTZ for drawing and Bravura's SMuFL metadata for music-glyph geometry.